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DeCoMon: Depth Completion-Guided Monocular Depth Estimation via Knowledge Distillation

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Seyoung Joung

2025

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이 논문을 석사학위 논문으로 제출함

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Summary(Abstract)

Monocular depth estimation (MDE) is a fundamental computer vision task crucial for applications in autonomous driving and robotics. However, accurately predicting absolute depth scale from monocular inputs alone is inherently challenging, significantly limiting its practical deployment. Although LiDAR-based depth completion (DC) techniques offer precise absolute depth measurements, they remain impractical for widespread use due to high cost and environmental constraints. In this paper, we propose an **Inverse-Depth Knowledge Distillation (KD)** framework, transferring accurate absolute depth information from a LiDAR-free adaptation of the DC teacher to a lightweight, self-supervised monocular MDE student.

Our **domain-gap-aware distillation** operates entirely in the inverse-depth domain. Concretely, we convert the teacher's metric depth to inverse depth and align the student's raw disparity predictions by applying an **image-wise median-scale normalization**. The resulting aligned target provides scale-consistent supervision without additional learnable parameters. By enforcing this median-scale alignment at every training iteration, we ensure that each component of our unified distillation loss—absolute scale alignment (L1-KD), scale-invariant structure preservation (SiLog-KD), and feature-level geometric consistency (AffinityKD)—operates in a depth-friendly representation that directly inherits the teacher's absolute scale, while preserving the efficiency of monocular inference.

Empirically, our distilled model significantly resolves the scale ambiguity inherent to monocular predictions, outperforming the baseline LiteMono by **on KITTI Eigen** and **Make3D**, thereby setting a new state-of-the-art performance among sub-10M-parameter monocular depth estimators.

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논 문 초 록

단안 깊이 추정(Monocular Depth Estimation, MDE)은 자율주행 및 로봇 응용을 위한 핵심 컴퓨터 비전 과제이다. 그러나 단일 RGB 입력만으로 정확한 절대 깊이 스케일을 예측하는 것은 본질적으로 매우 어렵기 때문에, 실제 시스템에 적용하는 데 근본적인 한계가 존재한다. LiDAR 기반의 깊이 보완(Depth Completion, DC) 모델은 매우 정밀한 절대 깊이를 제공할 수 있지만, LiDAR 센서의 높은 비용과 환경적 제약으로 인해 광범위한 실용화가 어렵다는 단점이 있다. 본 연구에서는 LiDAR 를 제거한 DC 교사 모델을 활용하여 정확한 절대 깊이 정보를 경량의 자기지도 기반 단안 깊이 추정 학생 모델에 전달하는 **역-깊이 지식 증류(Inverse-Depth Knowledge Distillation, KD)** 프레임워크를 제안한다.

본 연구에서 제안하는 도메인 차이 인지형 지식 증류(Domain-gap-aware Distillation)는 전적으로 역-깊이 공간에서 이루어진다. 구체적으로, 교사 모델의 미터 단위 깊이를 역-깊이로 변환하고, 학생 모델의 원시적인 시차(disparity) 예측값은 이미지 단위의 중앙값(median-scale)을 활용하여 같은 역-깊이 공간으로 정렬한다. 이를 통해 추가적인 학습 가능한 파라미터 없이 스케일 일관성(scale-consistency)을 갖는 감독 신호를 생성한다. 본 논문의 통합 지식 증류 손실을 구성하는 각각의 항목 절대 스케일 정렬(L1-KD), 스케일 불변의 상대적 구조 유지(SiLog-KD), 특징 수준의 구조적 유사성 유지(AffinityKD) 은 모두 이러한 이미지별 median-scale 정렬을 통해 깊이에 친화적인(depth-friendly) 표현 공간에서 동작하며, 교사의 절대 깊이 스케일을 학생 모델이 직접적으로 학습하도록 만든다. 이를 통해 단안 추론의 효율성을 유지하면서도 정확한 절대 스케일 정보를 학생 모델에 효과적으로 전수한다.

제안하는 방법은 실험적으로 단안 예측에서의 스케일 모호성을 현저히 개선했으며, 대표적인 벤치마크 데이터셋 KITTI Eigen 과 Make3D 에서 베이스라인 모델인 LiteMono 대비 성능 향상을 기록하였다. 이는 10M 이하의 경량 단안 깊이 추정 모델 중 새로운 최고 성능을 제시한 결과이다.

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I. Introduction

Monocular depth estimation (MDE) has emerged as a fundamental task in computer vision, with extensive applications in autonomous driving, robotics, and 3D reconstruction [1, 5]. Recent advances in deep learning have significantly improved the performance of MDE methods. These range from supervised approaches trained directly on accurate LiDAR ground-truth depth to self-supervised frameworks relying on geometric cues from videos or stereo pairs without explicit depth annotations [6, 7, 8]. Concurrently, knowledge distillation (KD) has gained prominence as a powerful technique to effectively transfer the performance and representation capabilities from large teacher networks into compact, efficient student models [2, 10, 14]. By doing so, KD enables lightweight student models to preserve high accuracy while being deployable in resource-constrained environments.

Knowledge distillation has been increasingly explored within the context of MDE to enhance model efficiency and performance. For instance, large-scale MDE models can be distilled into compact monocular networks where computational complexity is significantly reduced and accuracy while accuracy being maintained [2, 10, 14]. Moreover, recent works show that distilling geometric cues from stereo-trained teachers effectively mitigate the scale ambiguity inherent to MDE and enhance student model robustness across diverse conditions [20, 30]. However, despite these advancements, MDE methods remain fundamentally limited by scale ambiguity. Monocular cues alone are inherently insufficient to reliably infer absolute metric depth. Although KD from stereo teachers provides valuable geometric guidance, it is still limited in addressing occlusion issues and correspondence errors inherent in stereo vision systems [30]. Consequently, monocular

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models distilled from either monocular or stereo teachers remain limited in accurately capturing absolute depth scale, which works as a critical barrier to real-world applications.

To obtain accurate and reliable metric depth, depth completion (DC) techniques have been actively developed. DC combines RGB images with sparse LiDAR measurements to generate dense and precise depth maps [3, 9, 28, 28]. State-of-the-art DC networks, such as CompletionFormer [3] and GuideFormer [29], effectively integrate local and global contexts by combining convolutional and transformer architecture and have yielded exceptional performance on benchmarks like KITTI. Unlike MDE methods, DC inherently leverages explicit geometric information from LiDAR sensors. DC can resolve scale ambiguity and enhance depth estimation accuracy. However, despite their superior accuracy, DC approaches are limited by their dependence on costly depth sensors (e.g., LiDAR): LiDAR systems are expensive and require careful calibration. Furthermore, they often face limitations in availability and environmental constraints, which poses challenges for widespread practical adoption in purely vision-based applications.

To address these limitations, we introduce **DeCoMon**, a novel MDE framework guided by depth completion (DC) based on knowledge distillation. For the teacher model, we leverage a LiDAR-based DC network (CompletionFormer) [3]. To adapt the DC model to monocular input, we feed an RGB image with an all-zero (blank) depth channel. In this way, DC architecture can operate in a monocular mode while also maintaining its structural capacity. This teacher is trained with accurate LiDAR ground truth on datasets such as KITTI [27] and learns to predict absolute metric depth from a single image. High-fidelity output of this teacher provides dense and reliable predictions as pseudo-ground-truth for a lightweight RGB-only student model trained through distillation. For the student model, it is supervised by a scale-invariant regression loss against the teacher's output, and is further regularized via feature distillation and self-supervised photometric

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reconstruction as well as smoothness loss [6, 24]. Consequently, under this joint scheme, the student model benefits from both the teacher's LiDAR-based accuracy and classical monocular cues. Importantly, the student requires only a single RGB image at inference, which renders our approach broadly applicable.

DeCoMon is conceptually distinct from prior depth distillation approaches, which typically constrain teachers and students to the same input modality. Monocular-to-monocular methods, such as Zhou & Dong [17], rely primarily on self-distillation within single-image frameworks, while stereo-to-monocular distillation methods leverage stereo-derived geometric cues but still suffer from occlusion and correspondence errors [20, 30]. Additionally, conventional feature-based KD methods often impose architectural constraints, limiting flexibility [15, 16]. However, **DeCoMon** effectively bridges such gaps between DC (with LiDAR) and monocular estimation: this approach enables a student to acquire absolute scale and rich geometric structure from a teacher with privileged sensing. This framework robustly handles differences in teacher and student input domains via output normalization and intermediate feature distillation, achieving significantly improved absolute depth estimation and scale consistency on the KITTI benchmark [27] compared to the baseline student model.

Our key contributions are as follows:

1. **LiDAR-Guided Teacher–Student Framework.** We introduce **DeCoMon**, a teacher–student paradigm in which a LiDAR-trained depth-completion network serves as a high-capacity teacher and a lightweight monocular network acts as the student. The teacher supplies high-fidelity, metric-scale depth targets, while the student performs real-time inference from a single RGB image.

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2. **LiDAR-Free Adaptation of the Depth-Completion Teacher.** We repurpose the depth-completion network for RGB-only input by fine-tuning it **without LiDAR measurements**, preserving its ability to predict absolute depth. The resulting teacher delivers precise, metric supervision even though no LiDAR is available at test time.

3. **Domain-Gap-Aware Distillation.** To reconcile the teacher's metric depth with the student's relative disparity, we used **per-image median-scale alignment in inverse-depth space during** training. This normalization enables a unified loss comprising L1, SiLog, and AffinityKD, which eliminates scale ambiguity and markedly boosts student accuracy.

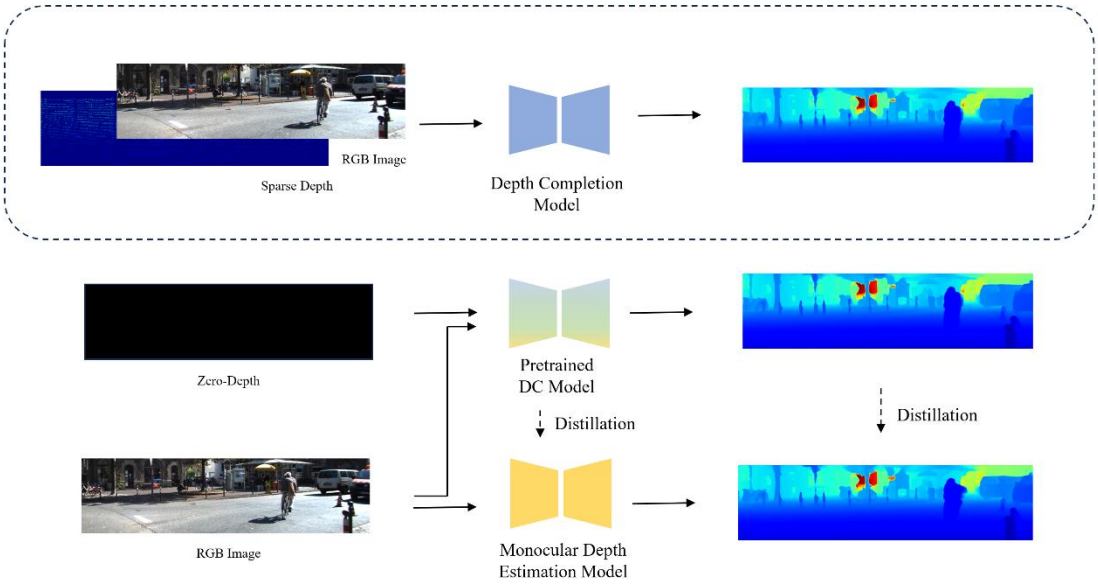

Figure 1. Overview of DeCoMon Architecture.
knowledge distillation from a depth completion teacher (CompletionFormer) to a monocular depth estimation student (Lite-Mono). The teacher receives RGB images combined with zero-depth map, and the student takes RGB-only inputs. Distillation is performed via feature-level affinity transfer and self-supervised regularizers, effectively addressing scale ambiguity inherent in monocular methods.

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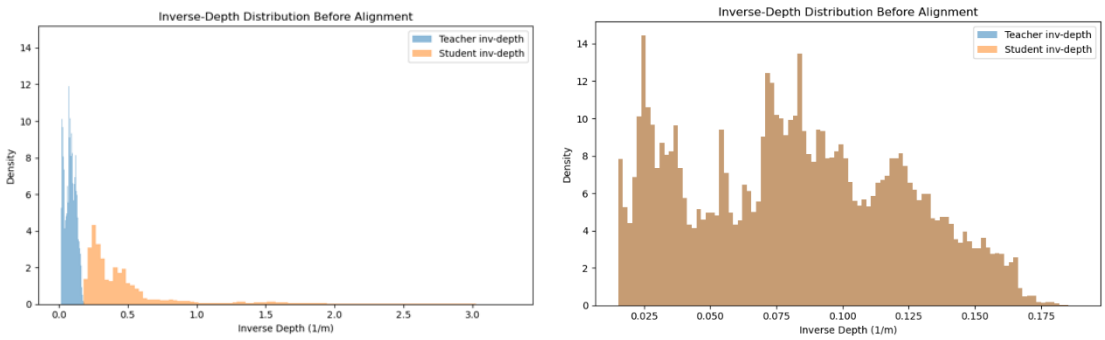

Figure 2. Inverse-Depth Distribution Misalignment Prior to Normalization

Distribution analysis of inverse-depth predictions from the teacher and student before alignment. **Left:** The teacher and student exhibit distinct distribution patterns, with the student biased toward smaller inverse-depth values, indicating a significant scale discrepancy. **Right:** A zoomed-in view of the student distribution reveals irregular density and noisy variance, motivating the need for normalization to enable stable and scale-consistent supervision.

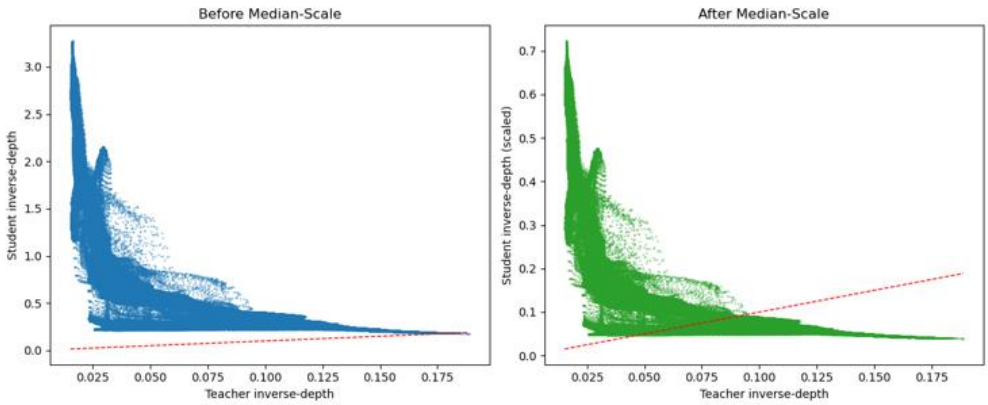

Figure 3. Inverse-depth scatter plots before and after median-scale alignment.

(Left) Without normalization, the student's inverse-depth values deviate significantly from the teacher's. (Right) After per-image median-scale alignment, student predictions align closely with the teacher along the identity line, enabling scale-consistent loss computation. In both plots, the red line indicates the ideal alignment line (y = x). The clear deviation from this reference highlights the necessity of median-scale normalization for stable, scale-consistent supervision.

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II. Related Works

1. Monocular Depth Estimation

MDE, a task of inferring pixel-wise scene distance from a single RGB image, has achieved notable progress as a regression approach based on deep learning [5, 6]. On autonomous-driving datasets such as KITTI [27], supervised approaches that leverage sparse LiDAR depth samples as ground truth have reached substantial accuracy. Self-supervised MDE methods, on the other hand, train without any ground truth depth annotations, utilizing photometric view-synthesis errors across stereo pairs of consecutive video frames [7, 8]. Despite these advances that contributed to higher performance, monocular input still faces fundamental limitations: scale ambiguity in absolute depth and difficulty reconstructing fine-grained structure. Training with sparse LiDAR ground truth alone has even been reported to introduce noises such as stripe-like artifacts in the predicted depth maps [18]. To address these issues, several recent studies have introduced dense teacher signals or knowledge distillation frameworks to help the monocular student network overcome limitations in scale ambiguity and structural detail [21, 22]. We address these issues by suggesting a novel MDE framework, **DeCoMon**, which distills knowledge from a Depth Completion pretrained model whose dense predictions serve as supervisory signals.

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2. Depth Completion

Depth Completion seeks to reconstruct dense depth maps by combining color images and sparse depth sensors such as LiDAR and has been actively researched for autonomous driving and robotics [3, 9, 28, 29]. Early methods typically relied on straightforward convolutional fusion between RGB images and sparse depth cues [9, 28]. Subsequent approaches significantly improved performance by explicitly modeling spatial relationships using propagation or affinity-based networks, as demonstrated by methods like Spatial Propagation Networks (SPN) [9] and Non-Local Spatial Propagation Networks (NLSPN) [28], which explicitly enforce spatial consistency through pixel-wise affinity. More recent advances have introduced transformer-based architectures such as GuideFormer [29] and CompletionFormer [3], which integrate convolutions and vision transformers to effectively capture both fine-grained local structures and long-range contextual information, achieving state-of-the-art performance on benchmarks such as KITTI [27]. Although these Depth Completion methods provide highly precise depth estimates, they still depend on depth-sensor input and typically employ large computational models, which limits their use in purely vision-based applications. **DeCoMon** overcomes such limitations by distilling knowledge from a high-accuracy depth-completion teacher into a monocular student network. This framework thus inherits the advantages of dense depth supervision without requiring any additional sensors.

3. Knowledge Distillation

Knowledge Distillation for depth estimation has been widely applied to transfer knowledge from high-capacity teachers to efficient student models in depth estimation [2, 10, 15, 16]. Studies on

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unsupervised MDE have adopted teacher-student frameworks that refine predictions and achieve high accuracy without additional depth labels [2]. Other work has successfully improved MDE performance on mobile devices by transferring the representational capacity of complex teacher networks to lightweight students [18]. Methods that leverage multi-scale feature fusion along with confidence-aware self-distillation were also proposed to improve depth prediction in far-range and occluded regions [20]. In addition, studies have shown that using spatial cues and uncertainty estimates from a stereo-based teacher can strengthen the 3D reasoning of monocular networks [21], while multi-teacher distillation methods that aggregate knowledge from several teachers enhance the zero-shot generalization performance of monocular models [22]. Recent efforts even aim to distill monocular models into depth-completion tasks to achieve high-resolution depth recovery [4]. In the current study, we propose a new novel perspective that treats a Depth Completion network as the teacher, **DeCoMon**, which transfers its sensor-level accuracy and dense depth cues to a monocular student. In this way, we aim to address and overcome existing limitations of MDE approaches.

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III. Methods

1. Overview

We suggest a teacher-student KD pipeline that leverages the expressive power of LiDAR-based DC networks [3]. Our teacher network is trained on RGB images with accurate LiDAR ground truth depth maps. During distillation, we feed the teacher network an RGB image along with a zero-valued depth channel (i.e., a blank depth map of the same spatial resolution) to preserve the original two-channel (RGB + depth) input structure while forcing the teacher to operate in a monocular setting. After three-stage curriculum fine-tuning detailed below under this condition, the teacher can produce reliable absolute metric depth predictions from a single image.

For the student model, a lightweight monocular network based on LiteMono [14] predicts depth in the disparity domain using only RGB input. Important is that the student is never supervised with real LiDAR data, it instead learns only via distillation from the teacher's predictions that serve as dense pseudo ground truth. We normalize both outputs from teacher and student to the inverse-depth space for comparison in order to address the domain gap and ensure the student captures both the absolute scale and the detailed geometric structure of the teacher's predictions. We specifically applied photometric reconstruction [6] for the student training. The training further regularized with self-supervised losses leads to minimized photometric inconsistency across views and to edge-aware smoothness [5, 6] to encourage locally coherent depth. By doing

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so, under this framework, the student inherits the rich scene understanding of the teacher while also remaining as a highly efficient and lightweight monocular depth model.

2. Teacher Network: Three-Stage Curriculum Fine-tuning

Our teacher is based on a LiDAR-supervised DC model (e.g., CompletionFormer [3]). As shown in Figure 1 and Table 1, the teacher is first pre-trained on the KITTI dataset using both RGB images and sparse LiDAR depth, then fine-tuned for monocular inference. Intermediate checkpoints are saved after each stage and released together with training scripts to ensure exact reproducibility. The fine-tuning procedure follows a three-stage curriculum outlined below:

Stage	Loss	Objective
A (Stability Warm-up)	$\mathcal{L}_1$	Preventing depth collapse
B (Scale Correction)	$\mathcal{L}_{1_{inv}}$	Scale stabilization & optimization
C (Detail Refinement)	$\mathcal{L}_{1_{inv}} + \mathcal{L}_{\text{BerHu}}$	Outlier suppression
	$\mathcal{L}_{1_{inv}} + \mathcal{L}_{\text{SiLog}} + \mathcal{L}_{\text{smooth}} + \mathcal{L}_{\text{BerHu}}$	Structural and boundary refinement

Table 1. Three-stage Fine-tuning of the Teacher Network (DC)

Stage A: Stability Warm-Up

The teacher initially receives an RGB image and a zero-valued depth map (of the same size as the RGB image) as inputs. As this drastic change may destabilize training, we start with masked $\mathcal{L}_1$ loss alone, ensuring a reasonable depth distribution and avoiding divergence due to extreme errors.

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$$\mathcal{L}_1 = \frac{1}{N}\sum_i M_i |d_i - d_i^*| \tag{1}$$

M_i is the valid mask, d_i is the predicted depth, d_i^* is ground truth, and N is the number of valid pixels. This stage ensures that the network stably learns the basic depth structure without collapse.

When a depth completion model receives only RGB inputs, gradient vanishing and depth collapse inevitably occur. To mitigate this issue, we freeze the conv_depth branch of the depth completion network (originally designed for LiDAR input) and input a zero-valued depth map instead. By computing the loss exclusively on valid pixels from the ground-truth and treating empty regions as zeros, the masked $\mathcal{L}_1$ is guided away from trivial zero-depth solutions, effectively preventing depth collapse.

Stage B: Scale Correction

After stabilizing, we apply inverse-depth normalization and a cosine learning rate schedule. Continued masked inverse $\mathcal{L}_1$ loss is used, but the model now regresses in the inverse-depth domain in order to align predicted scales to real-world metric depth. This process ensures minimizing global scale mismatches and retaining stable convergence. [10]

$$\mathcal{L}_{1_{inv}} = \frac{1}{N}\sum_i M_i |z_i - z_i^*| \tag{2}$$

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$z_i = 1/d_i$ is the predicted inverse-depth, $z_i^* = 1/d_i^*$ is inverse-ground truth. The depth distribution is constrained within a narrow and uniform range, ensuring stable and consistent scale throughout training. We pretrain teacher network with a cosine learning rate schedule to achieve smooth learning dynamics and facilitate prolonged, fine-grained updates.

Stage C: Detail Refinement

Stage C-1: BerHu Outlier Loss

In Stage C1 (Epochs 4–5), we introduce the masked BerHu loss [19] to improve robustness against large depth errors (outliers). BerHu [19], also known as the reverse Huber loss, is applied here to the inverse-depth differences between predictions and ground-truth values. Specifically, for each valid pixel i, we define the absolute inverse-depth error as $\delta_i = |z_i - z_i^*|$, where z_i and z_i^* denote the predicted and target inverse-depth values, respectively. The dynamic clamping threshold c is adaptively determined at the top 10% percentile of the error distribution within the current batch: $c = 0.2 \times percentile_{90}(\{\delta_i \mid M_i = 1\})$, where $M_i \in \{0,1\}$ indicates the validity mask for depth pixels. The BerHu penalty function $\phi(\delta)$ is defined such that errors smaller than or equal to the threshold c are penalized linearly, while larger errors (exceeding c) increase smoothly via a quadratic function:

$$\phi(\delta) = \begin{cases} \delta, & \delta < c \\ \dfrac{\delta^2 + c^2}{2c} \end{cases} \tag{3}$$

Thus, the BerHu loss [19] $\mathcal{L}_{BerHu}$ is defined as the mean of $\phi(\delta)$ across all valid pixels:

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$$\mathcal{L}_{BerHu} = \frac{1}{N}\sum_{i} M_i \phi(\delta_i) \tag{4}$$

Here, $N = \sum_i M_i$ denotes the number of valid pixels. In Stage C1, we optimize a weighted combination of BerHu [19] and L1 losses with a ratio of 0.3:0.7 (see Table 1). Incorporating the BerHu term mitigates the impact of large depth errors, rendering training more robust against outliers and significantly improving the model's accuracy.

$$\mathcal{L}_{\text{stage}C_1} = \lambda_{BerHu}\mathcal{L}_{\text{BerHu}} + \lambda_1 \mathcal{L}_1 \tag{5}$$

Stage C2: SiLog + Edge-aware Smoothness + BerHu Outlier Loss

After significantly reducing absolute depth errors in Stage C1, we further refine the relative depth structures and enhance local consistency in Stage C2 by introducing a scale-invariant logarithmic (SiLog) loss [1], an edge-aware smoothness loss [5] and BerHu loss [19]. The SiLog loss minimizes errors in logarithmic space between predicted and target depths, explicitly learning global scale-invariant depth relationships. Specifically, it is computed from the squared differences of log-depth values $s_i = ln\, z_i - ln\, z_i^*$, incorporating a term to compensate for global error biases (e.g., combining variance and squared mean with a coefficient $\lambda = 0.85$. Consequently, the absolute scale discrepancies are neglected, guiding the model to focus on relative depth proportions.

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$$\mathcal{L}_{SiLog} = \sqrt{\frac{1}{N}\sum_i M_i s_i^2 - \lambda\left(\frac{1}{N}\sum_i M_i s_i\right)^2}, \tag{6}$$

$$s_i = \ln z_i - \ln z_i^* \tag{7}$$

Meanwhile, the edge-aware smoothness loss acts as a regularizer that penalizes depth variations between neighboring pixels to promote local continuity. It is designed to assign stronger penalties in uniform regions (with small image gradients) while alleviating penalties near prominent edge boundaries.

$$\mathcal{L}_{\text{smooth}} = \frac{1}{N}\sum_i M_i \left(|\partial_x \boldsymbol{ln} z_i| e^{-\alpha|\partial_x I_i|} + |\partial_y \boldsymbol{ln} z_i| e^{-\alpha|\partial_y I_i|}\right), (\alpha = 10) \tag{8}$$

In Stage C2, we combine the SiLog [1], smoothness losses [5] and BerHu loss [19] along with a residual $\mathcal{L}_1$ loss into a composite loss function:

$$\mathcal{L}_{\text{stage}C_2} = \lambda_1 \mathcal{L}_1 + \lambda_{SiLog}\mathcal{L}_{SiLog} + \lambda_{smooth}\mathcal{L}_{Smooth} + \lambda_{BerHu}\mathcal{L}_{\text{BerHu}} \tag{9}$$

fine-tuning the model accordingly. Specifically, we continue using the BerHu loss [19] from Stage C1 but significantly reduce its weight, gradually shifting the focus toward fine-grained depth details. By progressively decreasing the influence of the BerHu term, we alleviate the strict

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penalization of large depth errors, allowing the SiLog loss [1] which emphasizes relative depth relationships to more effectively enhance local structural details. Consequently, Stage C2 substantially improves depth prediction quality, resolving subtle depth contours, clarifying fine-scale structural nuances, and distinctly delineating object boundaries.

Through this three-stage curriculum, the teacher can predict highly reliable, metric-scale depth from RGB alone.

3. Student Network: Distillation and Self-supervised Learning

The LiteMono-based student network [14] predicts inverse depth (disparity) from monocular RGB inputs using self-supervised learning, whereas our teacher network is a fully-supervised depth completion (DC) model trained with accurate LiDAR measurements to predict absolute metric depth. This inherent difference in supervision introduces a significant **scale discrepancy** between the teacher's absolute depth and the student's relative disparity, manifesting as scale ambiguity and drift in the student's predictions.

To address this domain gap, we propose **Inverse-Depth Knowledge Distillation (KD)**, which unifies the prediction spaces of the two networks. Specifically, we convert the teacher's metric depth to inverse-depth and align the student's disparity predictions using a **per-image median-scale normalization**. This allows direct and scale-consistent supervision without requiring additional learnable parameters.

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Let $P_i^S \in (0,1]$ denote the student's raw disparity output after sigmoid activation. To align with the teacher's inverse-depth output z_i^T, we compute the per-image median scale [6] ms as:

$$ms = \underset{\{i|M_i=1\}}{median} \frac{z_i^T}{P_i^S + 10^{-3}}, \tag{10}$$

where M_i indicates valid pixel locations (as determined by LiDAR visibility and evaluation mask). If no valid pixels exist, we default to $ms = 1$. The aligned student predictionz_i^S is then given by:

$$z_i^S = ms \cdot P_i^S \tag{11}$$

This simple yet effective image-wise normalization strategy robustly anchors the student's inverse-depth predictions to the teacher's scale, mitigating frame-wise drift and improving stability during early training. Unlike global learnable scale factors, our method is parameter-free and provides faster convergence when used in conjunction with a OneCycle learning rate schedule. We visualize this process in **Figure 2**, which shows the distribution mismatch between teacher and student inverse-depth outputs prior to alignment. The student exhibits compressed and noisy predictions, clearly deviating from the teacher's well-formed distribution. **Figure 3** further illustrates the impact of median-scale normalization: the left scatter plot shows the misalignment between teacher and student predictions before scaling, while the right plot demonstrates how our method tightly aligns predictions along the identity line (y = x), enabling effective and scale-consistent supervision.

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3.1. Loss Function Definition

Our knowledge distillation losses are comprehensively structured as follows, clearly defined using the notation from Section Ⅲ.2:

- **L1-KD:**

$$\mathcal{L}_{L1_{KD}} = \frac{1}{N}\sum_{i} M_i |{z_i}^S - {z_i}^T| \tag{12}$$

The index i denotes the one-dimensional (1-D) vectorized index of valid pixels, obtained by flattening the spatial coordinates $(u, v) \in 1, \dots, H \times 1, \dots, W$ within each image.

- **SiLog-KD:**

$$\mathcal{L}_{\mathrm{SiLog}_{KD}} = \sqrt{\frac{1}{N}\sum_{i} M_i {s_i}^2 - \lambda\left(\frac{1}{N}\sum_{i} M_i s_i\right)^2}, \tag{13}$$

$$s_i = ln\ {z_i}^S - ln\, {z_i}^T, \tag{14}$$

This preserves relative depth structures and mitigates scale drift by compensating for global scale shifts.

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- **AffinityKD:**

We perform structural alignment between the teacher and student networks at 1/4 resolution via intermediate feature maps. Specifically, we project their respective representations into a common 128-channel space using 1×1 convolutions:

$$G_T(F_T) \in \mathbb{R}^{128 \times H \times W}, \qquad G_S(F_S) \in \mathbb{R}^{128 \times H \times W} \tag{15}$$

Next, we compute pairwise affinity using normalized feature vectors to measure cosine similarity between spatial positions p and q:

$$A_{pq} = \langle \widehat{f^p}, \widehat{f^q} \rangle, \qquad where \quad \widehat{f^p} = \frac{f^p}{\|f^p\|_2}, \qquad \widehat{f^q} = \frac{f^q}{\|f^q\|_2} \tag{16}$$

where f^p and f^q are feature vectors at positions p and q, respectively. This yields affinity matrices A^T and A^S of size $(HW \times HW)$ for the teacher and student.

To transfer spatial layout and geometric structure from the teacher to the student, we minimize the difference between their affinity matrices using the following loss:

$$\mathcal{L}_{\mathrm{Aff}_{KD}} = \frac{1}{H^2W^2} \sum_{p=1}^{HW} \sum_{q=1}^{HW} \left| {A_{pq}}^T - {A_{pq}}^S \right| \tag{17}$$

Unlike pointwise feature regression, this affinity-based formulation effectively transfers mid-to-large scale structural context while being robust to domain shift and resolution differences. As each affinity entry captures relative relationships between spatial locations, the method enables the student to preserve the teacher's structural priors beyond absolute feature magnitudes. This approach draws inspiration from similarity-preserving knowledge distillation [48].

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- **Photometric Loss:**

  We define the photometric reconstruction loss [6] as:

$$\mathcal{L}_{photo} = min_{j\in-1,1}\left\|I^t - \hat{I}^{t\rightarrow t+j}\right\| \tag{18}$$

  where I^t denotes the input target frame at time t, and $\hat{I}^{t\rightarrow t+j}$ is the image synthesized by warping the source frame I^{t+j} into the target view using the estimated depth and pose. This loss encourages geometric consistency by penalizing photometric reprojection errors across temporally adjacent frames, thereby refining texture boundaries and capturing subtle depth variations.

- **Edge-aware Smoothness:**

  Penalizes excessive depth fluctuations in smooth regions, preserving sharp transitions at edges and refining prediction quality.

AffinityKD [48] is applied at H/4 resolution between the teacher encoder's transformer Block-1 and student encoder's Stage-0, effectively balancing detail preservation and noise suppression while minimizing domain discrepancies. This transformer block naturally combines local CNN features and global attention, making it structurally compatible with the student's initial encoder block. AffinityKD [48] employs cosine similarity calculations that are robust to differences in channel dimensions, facilitated efficiently via 1×1 convolutional projections.

We set the KD loss weights as $\lambda_1 : \lambda_2 : \lambda_3 = 60 : 15 : 20$, explicitly assigning higher importance to absolute depth errors (L1-KD), followed by relative structural accuracy (SiLog-KD) [1] and feature-level alignment (AffinityKD) [48]. The photometric [6] and edge-aware smoothness losses [5, 6] are weighted consistently with established best practices in self-supervised monocular depth estimation. Consequently, the overall loss function becomes:

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$$\mathcal{L}_{\text{total}} = \lambda_1 L_{L1\,KD} + \lambda_2 \mathcal{L}_{\text{SiLog KD}} + \lambda_3 \mathcal{L}_{\text{Aff KD}} + \alpha \mathcal{L}_{\text{photo}} + \beta \mathcal{L}_{\text{smooth}} \tag{19}$$

During the first two epochs, we perform a warm-up with adjusted weights:

- **Epochs 1–2**: $\lambda_1 = 25, \lambda_2 = 25, \lambda_3 = 10$ with a photometric weight gradually increased linearly from 0 to 0.05 over the first 500 iterations and a smoothness weight of 0.10.
- **Epoch ≥ 3**: steady-state training with $\lambda_1 = 60, \lambda_2 = 15, \lambda_3 = 20$ photometric weight of 0.05, and smoothness weight of 0.20.

Training Outcomes

This meticulously structured KD loss framework ensures:

1. **Absolute Scale Alignment** is achieved via L1 and SiLog KD losses [1], effectively resolving the inherent scale ambiguity in monocular depth predictions.
2. **Structural Knowledge Transfer** is facilitated by AffinityKD [48], enabling robust learning of global layout and structural relationships independent of scale factors.
3. **Fine-Grained Detail Refinement** is accomplished through complementary photometric [6] and edge-aware smoothness losses [5, 6], resulting in improved texture boundaries and depth smoothness.

The design of these losses and normalization strategies robustly supports the claim that the teacher's absolute depth information is pivotal in practically resolving the student model's scale ambiguity. This logical and systematic approach is corroborated by empirical results, demonstrating consistent and meaningful improvements in absolute relative error. The resulting improvements in absolute relative error highlight the efficacy of our structured distillation framework in resolving monocular scale ambiguities and enhancing depth prediction accuracy

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IV. Results

1. Datasets

KITTI

We evaluate our method on the standard KITTI Eigen split [31], a widely adopted benchmark containing RGB images paired with sparse LiDAR-based ground-truth depth maps. The Eigen split provides 22,600 training images from 22 scenes and 697 test images from 28 scenes, with sparse depth labels covering roughly 5–6% of pixels per image [6, 23, 25]. We follow the standard evaluation protocol, employing the official KITTI evaluation script to mask invalid and occluded pixels, and compute standard metrics including RMSE, Absolute Relative Error (Abs Rel), and δ accuracy thresholds (δ_1, δ_2, δ_3) [6, 26].

Make3D

We evaluate zero-shot generalization performance on the Make3D dataset [37], which contains diverse outdoor scenes and represents a significant domain shift from KITTI [31]. Following prior works, we perform zero-shot inference without additional training or fine-tuning to test our method's robustness and generalization to novel scenarios.

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2. Implementation Details

For our knowledge distillation experiments, we adopt **Lite-Mono (3.1M parameters)** [14] as the student network and employ a pretrained depth-completion teacher network trained on RGB-only input. Both teacher and student models utilize an identical preprocessing pipeline, resizing input images to a resolution of 1024 ×320. Specifically, the AffinityKD component transfers structural knowledge by computing the feature affinity between teacher and student at ¼ resolution (with channel dimension set to 128). The models are optimized using the AdamW optimizer [32] (β = 0.9/0.98, weight decay = 1e-4) and trained with a OneCycleLR schedule [33], where the learning rate starts at 3e-4, peaks at 6e-4 (with pct_start set to 0.3), and then anneals down to 3e-5 over 3 epochs. To ensure training stability, we employ gradient clipping (clip value = 1.0) and utilize mixed-precision training (fp16 mode) for computational efficiency.

For our knowledge distillation experiments, we adopt **Lite-Mono (3.1M parameters)** [14] as our lightweight student network and employ a pretrained RGB-only depth-completion network as our teacher. Both teacher and student models share an identical preprocessing pipeline, resizing input images to a resolution of **1024×320**. The **AffinityKD** component is applied at ¼ resolution features (128 channels) to transfer structural knowledge from the teacher to the student effectively. All models are optimized using **AdamW** [32] (β_1=0.9, β_2=0.999, weight decay=1e-5) and trained with a batch size of 8 on a single **NVIDIA Titan RTX 3090 GPU**. The training pipeline involves three stages with distinct learning rate schedulers:

1. Initial distillation employs a **OneCycleLR scheduler** [33] (initial_lr=3×10^{-4}, max_lr=6×10^{-4}, pct_start=0.3, anneal_strategy=cos). (Stage A & B)
2. Stages C utilize a **CosineAnnealingLR scheduler** [38] (initial_lr=1×10^{-4} for fine-tuning and 5×10^{-5} for micro-tuning), progressively decreasing the learning rate down to 1×10^{-5} and 5×10^{-6} respectively, ensuring stable and fine-grained convergence.

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Gradient clipping (clip value=1.0) and mixed-precision training (fp16 mode) are additionally used to stabilize training and accelerate convergence.

3. KITTI Results

3.1 Teacher Fine-tuning Results

We fine-tuned our depth-completion teacher on the standard **KITTI Eigen split** at a resolution of **1024×320 pixels** through a structured three-stage curriculum, progressively enhancing its performance (Table 2). Initially, in Stage A, employing ground-truth masking significantly stabilized the model, improving the Absolute Relative error (Abs Rel) dramatically from 0.458 (near-collapse) down to 0.123, thereby establishing a robust baseline. In Stage B, converting depth predictions into inverse-depth representation further improved accuracy, achieving an Abs Rel of 0.118 and slightly increasing δ_1 accuracy to 0.862.

Subsequently, in Stage C, we introduced robust loss combinations to handle depth outliers and refine fine-grained structures. Specifically, combining L1 with BerHu loss [19] (0.7 L1 + 0.3 BerHu) notably reduced Abs Rel from 0.118 to 0.110. Finally, incorporating SiLog [1] and edge-aware smoothness terms (0.3 L1 + 0.4 SiLog + 0.2 Smooth + 0.1 BerHu) further refined local structural details and achieved the best overall results: an Abs Rel of 0.107, Sq Rel of 0.608, RMSE of 4.065, and a δ_1 accuracy of 0.880. This systematic fine-tuning scheme on KITTI effectively improved depth prediction quality and robustness for subsequent knowledge distillation tasks.

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3.2 Experimental Results and Comparisons

Table 3 reports monocular depth estimation results on the KITTI Eigen benchmark [31], comparing our proposed DeCoMon with a comprehensive set of recent state-of-the-art methods. Evaluations were conducted at both 640×192 and 1024×320 input resolutions.

At 640×192, our distilled DeCoMon model achieves an Abs Rel of 0.101, outperforming other lightweight baselines such as LiteMono [14] (0.107) and CADepth-Net [40] (0.105), while remaining competitive with larger-capacity models like Monodepth2 [6] and HR-Depth [35].

At the higher 1024×320 resolution, DeCoMon further improves to an Abs Rel of 0.093, achieving a Sq Rel of 0.539, RMSE of 3.751, and the highest δ_3 accuracy of 0.991 among all compared methods under 10M parameters. Notably, DeCoMon consistently outperforms its backbone baseline, LiteMono (Abs Rel: 0.103), by a substantial margin across all metrics.

These gains validate the effectiveness of our inverse-depth distillation strategy in bridging the domain gap between depth completion and monocular estimation. Despite using a compact 3.1M-parameter model, DeCoMon attains competitive performance against larger networks, demonstrating strong generalization while maintaining computational efficiency.

3.3 Qualitative Results and Analysis

Figure 4 presents qualitative comparisons between the baseline LiteMono [14] and our proposed **DeCoMon** across various driving scenes in the KITTI dataset. Overall, DeCoMon produces more

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structurally coherent and perceptually consistent depth maps, especially in terms of preserving geometric boundaries and long-range depth gradients.

Across multiple examples, DeCoMon demonstrates superior delineation of thin vertical structures such as poles, tree trunks, and traffic signs, while also improving the overall depth sharpness of foreground objects. Notably, background elements—particularly distant trees, buildings, and roads—exhibit smoother depth transitions and better atmospheric gradient handling, leading to more natural far-field representations. The student model benefits from the inverse-depth distillation, learning to suppress spurious responses caused by lighting, specularities, or textureless regions that LiteMono [14] often fails to ignore.

Interestingly, in several cases, LiteMono [14] still exhibits certain strengths. As shown in the third and fourth columns, LiteMono [14] produces flatter and more uniform road surfaces, avoids over-segmenting object silhouettes (e.g., rear vehicles), and better preserves subtle ground-plane gradations without excessive edge sharpening. These observations suggest that while DeCoMon excels at depth structure and geometric contrast, its stronger emphasis on global supervision may occasionally suppress delicate local continuity.

In summary, DeCoMon yields clear improvements in structural fidelity and long-range gradient modeling, outperforming LiteMono [14] in most scenarios. Nonetheless, we acknowledge remaining limitations, such as occasional overemphasis of depth boundaries and slight degradation in uniform surface smoothness. These findings highlight valuable directions for future refinement of distillation dynamics and scale-adaptive losses.

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Stage	Details	Result						
		Abs Rel (↓)	Sq Rel (↓)	RMSE (↓)	RMSE log (↓)	$\delta < 1.25$ (↑)	$\delta < 1.25^2$ (↑)	$\delta < 1.25^3$ (↑)
A	Teacher baseline	0.458	5.109	12.395	0.611	0.275	0.552	0.755
	GT masking	0.123	0.741	4.352	0.185	0.855	0.959	0.985
B	Inverse-depth	0.118	0.737	4.458	0.183	0.862	0.960	0.985
C	L1 + BerHu	0.110	0.662	4.393	0.175	0.870	0.964	0.987
	L1 + BerHu + SiLog + smooth	0.107	0.608	4.065	0.166	0.880	0.968	0.989

Table 2. Teacher Fine-tuning Results on KITTI Eigen Splits Quantitative results of the teacher model fine-tuned on the KITTI Eigen split. Each stage progressively improves depth prediction accuracy, with the final loss combination (L1 + BerHu + SiLog + smoothness) achieving the best overall performance across all metrics.

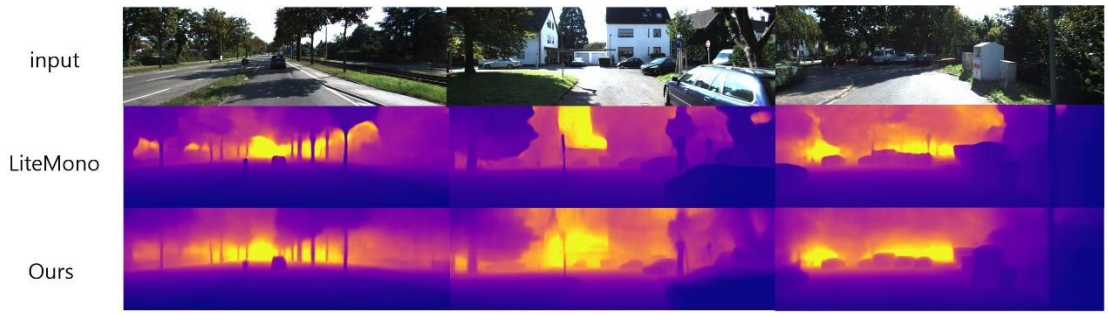

Figure 4. Qualitative Results on KITTI The top row shows the input images, the middle row presents depth predictions from the baseline LiteMono student model **[14]**, and the bottom row illustrates depth predictions from our proposed DeCoMon framework.

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Method	HxW	Abs Rel (↓)	Sq Rel (↓)	RMSE (↓)	RMSE log (↓)	δ < 1.25 (↑)	δ < 1.25^2 (↑)	δ < 1.25^3 (↑)	Params
PackNet-SfM [39]	640 x 192	0.111	0.785	4.601	0.189	0.878	0.960	0.982	128.3M
HR-Depth [35]	640 x 192	0.107	0.785	4.612	0.185	0.887	0.962	0.982	14.7M
Monodepth2 [6]	640 x 192	0.106	0.818	4.750	0.196	0.874	0.957	0.979	14.8M
Wang et al. [39]	640 x 192	0.106	0.799	4.662	0.187	0.889	0.961	0.982	16.8M
LiteMono [14]	640 x 192	0.107	0.765	4.561	0.183	0.886	0.963	0.983	3.1M
CADepth-Net [40]	640 x 192	0.105	0.769	4.535	0.181	0.892	0.964	0.983	58.3M
DynamicDepth [41]	640 x 192	0.096	0.720	4.458	0.175	0.897	0.964	0.984	36.4M
ManyDepth (MR, 36M) [42]	640 x 192	0.090	0.713	4.261	0.170	0.914	0.966	0.983	36.0M
SQLdepth (ResNet-50, 31M) [43]	640 x 192	**0.088**	0.697	4.175	0.167	**0.919**	**0.969**	0.984	31.2M
DeCoMon (ours)	640 x 192	0.101	**0.597**	**4.052**	**0.164**	0.877	0.967	**0.989**	3.1M
Monodepth2 [6]	1024 × 320	0.106	0.806	4.630	0.193	0.876	0.958	0.980	34.3M
Wang et al. [39]	1024 × 320	0.106	0.773	4.491	0.185	0.890	0.962	0.982	16.8M
LiteMono [14]	1024 × 320	0.103	0.757	4.449	0.180	0.894	0.964	0.983	3.1M
HR-Depth [35]	1024 × 320	0.101	0.716	4.395	0.179	0.899	0.966	0.983	14.7M
FeatDepth-MS [44]	1024 × 320	0.099	0.697	4.427	0.184	0.889	0.963	0.982	76.4M
DIFFNet [45]	1024 × 320	0.097	0.722	4.345	0.174	0.907	0.967	0.984	10.8M
CADepth-Net [40]	1024 × 320	0.096	0.694	4.264	0.173	0.908	0.968	0.984	58.3M
EPCDepth (ResNet-50) [46]	1024 × 320	0.091	0.646	4.207	0.176	0.901	0.966	0.983	50.6M
ManyDepth (ResNet-50, 37M) [42]	1024 × 320	0.087	0.685	4.142	0.167	0.920	0.968	0.983	37.4M
SQLdepth (ResNet-50, 37M) [43]	1024 × 320	**0.082**	0.607	3.914	0.160	**0.928**	**0.972**	0.985	37.3M
DeCoMon (ours)	1024 × 320	0.093	**0.539**	**3.751**	**0.154**	0.894	**0.972**	**0.991**	3.1M

Table 3. Experimental Results and Comparisons on KITTI Eigen Splits

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Method	Abs Rel (↓)	Sq Rel (↓)	RMSE (↓)	Log10 (↓)
Monodepth [5]	0.544	10.94	11.760	0.193
Zhou [7]	0.383	5.321	10.470	0.478
DDVO [47]	0.387	4.720	8.090	0.204
Monodepth2 [6]	0.322	3.589	7.417	0.163
CADepthNet [40]	0.312	3.086	7.066	0.159
LiteMono [14]	**0.305**	3.060	6.981	0.158
DeCoMon (ours)	0.307	**2.681**	**6.624**	**0.157**

Table 4. Experimental Results and Comparisons on Make3D

4. Make3D Results

Table 4 reports zero-shot generalization results on the Make3D dataset [37], where models are evaluated without any fine-tuning, using KITTI-pretrained weights. Following standard protocol, all methods are tested on the official test set with a center crop of 2:1 aspect ratio. Despite being trained solely on KITTI [31], our proposed **DeCoMon** achieves strong cross-dataset generalization, outperforming existing lightweight methods such as LiteMono [14] and CADepthNet [40] in terms of **Sq Rel**, **RMSE**, and **Log10 error**. These results demonstrate the robustness of our inverse-depth distillation strategy in transferring geometric priors across domains.

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5. Ablation Study

5.1 Ablation Study on Loss Combinations for Teacher Fine-tuning

Table 5 presents an ablation study on loss combinations used during teacher fine-tuning, highlighting the importance of proper loss scheduling. In Stage B, training with L1 (inverse-depth) serves as a stable baseline, yielding an Abs Rel of 0.118. Introducing SiLog loss prematurely at this stage (C#: L1 + SiLog) significantly degrades performance across all metrics (Abs Rel: 0.110), indicating that SiLog's scale-invariant nature is highly sensitive to unstable absolute predictions early in training.

In contrast, applying a robust BerHu loss formulation first (C1: L1 + BerHu) effectively reduces outliers and improves accuracy (Abs Rel: 0.110), establishing a more reliable foundation for scale refinement. Building on this, the final configuration (C2), which combines L1, BerHu, SiLog, and edge-aware smoothness, further refines depth structure and leads to the best overall performance (Abs Rel: 0.107, δ_1: 0.880). These results underscore that delaying the introduction of SiLog loss until after absolute-scale stabilization is crucial. A carefully staged loss design starting with L1, transitioning through BerHu [19], and incorporating SiLog [1] only after sufficient convergence enables optimal supervision for depth completion networks.

5.2 Impact of Depth Input Branch on Teacher Fine-tuning

Table 6 presents an ablation study evaluating the role of the depth input branch (conv_depth) in our depth-completion teacher architecture. In the RGB-only configuration (C2), where

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conv_depth is frozen and receives zero-valued input, the model achieves the best overall performance (Abs Rel: 0.107), validating that reliable monocular cues alone can drive accurate depth prediction. In contrast, supplying dummy depth input (mean=5m, with noise) to the unfrozen branch (C2#) results in degraded accuracy (Abs Rel: 0.111), likely due to unstable depth features introducing noise into early-stage representations. These findings confirm that freezing conv_depth and treating the network as an RGB-only model yields more stable and accurate convergence during teacher fine-tuning.

Stage	Loss	Abs Rel (↓)	Sq Rel (↓)	RMSE (↓)	RMSE log (↓)	$\delta < 1.25$ (↑)	$\delta < 1.25^2$ (↑)	$\delta < 1.25^3$ (↑)
B	L1(inv)	0.118	0.737	4.458	0.183	0.862	0.960	0.985
C#	L1 + SiLog	0.121	0.947	4.549	0.202	0.892	0.945	0.973
C1	L1 + BerHu	0.110	0.662	4.393	0.175	0.870	0.964	0.987
C2	L1 + BerHu + SiLog + + smooth	0.107	0.608	4.065	0.166	0.880	0.968	0.989

Table 5 Ablation Results of Teacher Finetuning 1 Ablation results for different loss combinations during teacher fine-tuning, demonstrating that introducing **BerHu loss prior to SiLog** yields superior performance, confirming that the ordering of loss integration critically impacts convergence and final accuracy.

Stage		Abs Rel (↓)	Sq Rel (↓)	RMSE (↓)	RMSE log (↓)	$\delta < 1.25$ (↑)	$\delta < 1.25^2$ (↑)	$\delta < 1.25^3$ (↑)
C2	**conv_depth 0**	0.107	0.608	4.065	0.166	0.880	0.968	0.989
C2#	**dummy depth**	0.111	0.635	4.214	0.176	0.866	0.949	0.988

Table 6 Ablation Results of Teacher Finetuning 2 Freezing the conv_depth branch with zero input (RGB-only) achieves superior performance, while activating it with noisy dummy depth degrades accuracy, highlighting the importance of stable depth input handling.

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V. Conclusion

In this paper, we presented **DeCoMon**, a novel inverse-depth knowledge distillation framework for monocular depth estimation (MDE). By transferring absolute metric depth from a LiDAR-supervised depth completion teacher to a self-supervised monocular student, DeCoMon effectively mitigates the scale ambiguity prevalent in monocular methods. Our approach introduces image-wise median scaling for stable output alignment, and integrates complementary distillation losses—including absolute depth (L1-KD), scale-invariant structure (SiLog-KD), structural affinity (AffinityKD), photometric, and edge-aware smoothness constraints. Extensive experiments on the KITTI dataset demonstrate that DeCoMon achieves state-of-the-art performance among lightweight MDE models. A key limitation lies in the dependency on high-quality depth completion teachers, which require accurate LiDAR data and precise sensor calibration. Nonetheless, when such a teacher is available, DeCoMon provides a robust and effective framework for transferring high-fidelity depth knowledge to compact monocular models.

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References

[1] D. Eigen, C. Puhrsch, and R. Fergus, “Depth Map Prediction from a Single Image using a Multi-Scale Deep Network,” in *Proc. NeurIPS*, 2014.

[2] A. Pilzer, S. Lathuilière, N. Sebe, and E. Ricci, “Refine and Distill: Exploiting Cycle-Inconsistency and Knowledge Distillation for Unsupervised Monocular Depth Estimation,” in *Proc. CVPR*, 2019.

[3] Y. Zhang, X. Sun, Y. Wang, C. Zhang, and Y. Shi, “CompletionFormer: Depth Completion with Convolutions and Vision Transformers,” in *Proc. CVPR*, 2023.

[4] Anonymous, “Distilling Monocular Foundation Model for Fine-grained Depth Completion,” in *Proc. CVPR*, 2025 (Poster).

[5] C. Godard, O. Mac Aodha, and G. J. Brostow, “Unsupervised Monocular Depth Estimation with Left-Right Consistency,” in *Proc. CVPR*, 2017.

[6] C. Godard, O. Mac Aodha, M. Firman, and G. J. Brostow, “Digging Into Self-Supervised Monocular Depth Estimation,” in *Proc. ICCV*, 2019.

[7] T. Zhou, M. Brown, N. Snavely, and D. G. Lowe, “Unsupervised Learning of Depth and Ego-Motion from Video,” in *Proc. CVPR*, 2017.

[8] J. Watson, M. Firman, G. Brostow, and D. Turmukhambetov, “Self-Supervised Monocular Depth Hints,” in *Proc. ICCV*, 2019.

[9] J. Uhrig, N. Schneider, L. Schneider, U. Franke, T. Brox, and A. Geiger, “Sparsity Invariant CNNs,” in *Proc. 3DV*, 2017.

[10] Y. Wang, W. Zhao, Z. Liu, J. Fu, and T. Mei, “Knowledge Distillation for Fast and Accurate Monocular Depth Estimation on Mobile Devices,” in *Proc. CVPR Workshops*, 2021.

[11] I. Alhashim and P. Wonka, “High Quality Monocular Depth Estimation via Transfer Learning,” *arXiv preprint arXiv:1812.11941*, 2018.

Collection @ ewha

[12] V. Casser, S. Pirk, R. Mahjourian, and A. Angelova, “Depth Prediction Without the Sensors: Leveraging Structure for Unsupervised Learning from Monocular Videos,” in *Proc. AAAI*, 2019.

[13] L. Fang, Z. Lyu, Y. Zhang, Y. Li, and L. Zhang, “Towards Good Practice for CNN-Based Monocular Depth Estimation,” in *Proc. WACV*, 2020.

[14] Z. Zhang, H. Liu, L. Wang, Y. Li, and G. Lin, “Lite-Mono: A Lightweight CNN and Transformer Architecture for Self-Supervised Monocular Depth Estimation,” in *Proc. CVPR*, 2023.

[15] A. Romero, N. Ballas, S. E. Kahou, A. Chassang, C. Gatta, and Y. Bengio, “FitNets: Hints for Thin Deep Nets,” in *Proc. ICLR*, 2015.

[16] S. Zagoruyko and N. Komodakis, “Paying More Attention to Attention: Improving the Performance of Convolutional Neural Networks via Attention Transfer,” in *Proc. ICLR*, 2017.

[17] Z. Zhou and B. Dong, “Self-distilled Feature Aggregation for Self-supervised Monocular Depth Estimation,” in *Proc. ECCV*, 2022.

[18] B. Choi, J. Lee, D. Kim, and J. Park, “Monocular Depth Estimation from a Fisheye Camera Based on Knowledge Distillation,” *MDPI Sensors*, 2023.

[19] A. B. Owen, “A Robust Hybrid of Lasso and Ridge Regression,” *Contemp. Math.*, vol. 443(7), pp. 59–72, 2007.

[20] Y. Wu, J. Xu, W. Zhu, and Z. Lin, “ADU-Depth: Attention-based Distillation with Uncertainty Modeling for Depth Estimation,” in *Proc. CoRL*, 2023.

[21] T. He, L. Zheng, J. Li, and J. Liu, “Distill-Any-Depth: Distillation Creates a Stronger Monocular Depth Estimator,” *arXiv preprint arXiv:2403.11392*, 2024.

[22] Q. Li, X. Feng, M. Liu, and C. Xie, “Cross-Modal Knowledge Distillation for Depth-Privileged Monocular Visual Odometry,” in *IEEE Robot. Autom. Lett.*, 2022.

[23] D. Eigen, C. Puhrsch, and R. Fergus, “Depth Map Prediction from a Single Image using a Multi-Scale Deep Network,” in *Proc. NeurIPS*, 2014.

Collection @ ewha

[24] H. Zhou, D. Greenwood, and S. Taylor, “Self-supervised Monocular Depth Estimation with Internal Feature Fusion,” *arXiv preprint arXiv:2110.09482*, 2021.

[25] R. Ranftl, K. Lasinger, D. Hafner, K. Schindler, and V. Koltun, “Towards Robust Monocular Depth Estimation: Mixing Datasets for Zero-shot Cross-dataset Transfer,” *IEEE Trans. Pattern Anal. Mach. Intell.*, 2022.

[26] H. Fu, M. Gong, C. Wang, K. Batmanghelich, and D. Tao, “Deep Ordinal Regression Network for Monocular Depth Estimation,” in *Proc. CVPR*, 2018.

[27] A. Geiger, P. Lenz, C. Stiller, and R. Urtasun, “Vision Meets Robotics: The KITTI Dataset,” *Int. J. Robot. Res.*, vol. 32(11), pp. 1231–1237, 2013.

[28] J. Park, K. Jeong, S. Joo, and K. M. Lee, “Non-Local Spatial Propagation Network for Depth Completion,” in *Proc. ECCV*, 2020.

[29] J. Rho, J. Park, H. Lee, and I. S. Kweon, “GuideFormer: Transformers for Image Guided Depth Completion,” in *Proc. CVPR*, 2022.

[30] Y. Guo, G. Li, N. Xu, and J. Ren, “Learning Monocular Depth by Distilling Cross-Domain Stereo Networks,” in *Proc. ECCV*, 2018.

[31] D. Eigen and R. Fergus, “Predicting Depth, Surface Normals and Semantic Labels with a Common Multi-Scale Convolutional Architecture,” in *Proc. ICCV*, 2015.

[32] I. Loshchilov and F. Hutter, “Decoupled Weight Decay Regularization,” in *Proc. ICLR*, 2018.

[33] L. N. Smith and N. Topin, “Super-convergence: Very Fast Training of Neural Networks Using Large Learning Rates,” in *Proc. SPIE: AI and ML for Multi-Domain Operations Apps*, vol. 11006, pp. 369–386, 2019.

[34] M. Klingner, J.-A. Termöhlen, J. Mikolajczyk, and T. Fingscheidt, “Self-Supervised Monocular Depth Estimation: Solving the Dynamic Object Problem by Semantic Guidance,” in *Proc. ECCV*, 2020.

Collection @ ewha

[35] X. Lyu, L. Liu, M. Wang, X. Kong, L. Liu, Y. Liu, X. Chen, and Y. Yuan, "HR-Depth: High Resolution Self-Supervised Monocular Depth Estimation," in *Proc. AAAI*, 2021.

[36] J. Bae, S. Moon, and S. Im, "MonoFormer: Towards Generalization of Self-Supervised Monocular Depth Estimation with Transformers," *arXiv preprint arXiv:2205.11083*, 2022.

[37] A. Saxena, M. Sun, and A. Y. Ng, "Make3D: Learning 3D Scene Structure from a Single Still Image," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31(5), pp. 824–840, 2008.

[38] I. Loshchilov and F. Hutter, "SGDR: Stochastic Gradient Descent with Warm Restarts," in *Proc. ICLR*, 2017.

[39] J. Wang, G. Zhang, Z. Wu, X. Li, and L. Liu, "Self-Supervised Joint Learning Framework of Depth Estimation via Implicit Cues," *arXiv preprint arXiv:2006.09876*, 2020.

[40] J. Yan, H. Zhao, P. Bu, and Y. Jin, "Channel-Wise Attention-Based Network for Self-Supervised Monocular Depth Estimation," in *Proc. 3DV*, 2021.

[41] Z. Feng, L. Yang, L. Jing, H. Wang, Y. Tian, and B. Li, "Disentangling Object Motion and Occlusion for Unsupervised Multi-Frame Monocular Depth," *arXiv preprint arXiv:2203.15174*, 2022.

[42] J. Watson, O. Mac Aodha, V. Prisacariu, G. Brostow, and M. Firman, "The Temporal Opportunist: Self-Supervised Multi-Frame Monocular Depth," in *Proc. CVPR*, 2021.

[43] Y. Wang, X. Liu, Y. Wang, X. Xu, and M. Yang, "SQLdepth: Generalizable Self-Supervised Fine-Structured Monocular Depth Estimation," in *Proc. AAAI*, 2024.

[44] C. Shu, K. Yu, Z. Duan, and K. Yang, "Feature-Metric Loss for Self-Supervised Learning of Depth and Egomotion," in *Proc. ECCV*, 2020.

[45] H. Zhou, D. Greenwood, and S. Taylor, "Self-Supervised Monocular Depth Estimation with Internal Feature Fusion," *arXiv preprint arXiv:2110.09482*, 2021.

Collection @ ewha

[46] R. Peng, R. Wang, Y. Lai, L. Tang, and Y. Cai, “Excavating the Potential Capacity of Self-Supervised Monocular Depth Estimation,” in *Proc. ICCV*, 2021.

[47] C. Wang, J. M. Buenaposada, R. Zhu, and S. Lucey, “Learning Depth from Monocular Videos Using Direct Methods,” in *Proc. CVPR*, 2017.

[48] F. Tung and G. Mori, “Similarity-Preserving Knowledge Distillation,” in *Proc. ICCV*, 2019.

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